# Machine Learning Course Project Guide

The project titles listed in this guide are intended solely as illustrative examples. Students are expected to select a project topic that is relevant to the course material while also reflecting their own academic and research interests. Proposals that differ from the suggested topics are welcome, provided they demonstrate a clear connection to machine learning concepts covered in the course.

You are encouraged to explore topics that match both the course themes and your personal interests. You may choose one of the suggested projects or propose your own, as long as it connects to machine learning and fits the goals of the course. If you have an idea outside this list, feel free to discuss it with the instructor.

## Requirements

- **Proposal (2-3 pages):** Problem statement, dataset, methods, timeline.
- **Presentation (15 mins / time):** Demo + key findings.
- **Midterm Report (4–5 pages):** Preliminary results + adjustments.
- **Final Report (6–8 pages):** Full pipeline, results, evaluation, discussion.

---

## Grading Criteria

- **Project Proposal:** 20%
- **Midterm Report:** 20%
- **Implementation & Results:** 30%
- **Final Report Quality:** 30%

## Potential ML Project Proposal

## The projects can be proposed in some directions:

- **ML foundation models: RL/IL/LLM/VLM/AVM**
- **ML for general human-machine interaction or action-vison-language task**
- **ML for Robotics ***
- **ML for Agent (Gaming, navigation, etc.)**
- **ML for Quantitative Computing * ***
- **ML in trustworthy infrastructure and safety system**
- **……..**

---

## **A list of 40 machine learning course projects** across the 6 directions.

This list covers:

- **7 projects in foundation models**
- **7 in human-machine interaction**
- **7 in robotics**
- **7 in agent-based systems**
- **7 in quantitative computing**
- **5 in trustworthy infrastructure**

Each project will include:

- **Title**
- **Dataset**
- **Method & Methodology (ML and AI methods)**
- **Possible Datasets**

---

**I. ML Foundation Models (RL / IL / LLM / VLM / AVM)**

1. **Fine-Tuning LLMs for Scientific Q&A**
   - Dataset: PubMedQA, SQuAD
   - Method: LoRA fine-tuning of GPT/LLaMA with transfer learning
2. **Visual-Language Model for Image Captioning**
   - Dataset: MS COCO

    - Method: CLIP + Transformer decoder for caption generation
3. **Autonomous Video Modeling (AVM) for Human Action Recognition**
    - Dataset: Kinetics-700
    - Method: Video Transformer (TimeSformer, ViViT)
4. **Imitation Learning from Human Demonstrations**
    - Dataset: D4RL (Hopper, HalfCheetah)
    - Method: Behavior Cloning + GAIL (Generative Adversarial Imitation Learning)
5. **Reinforcement Learning with Pretrained Foundation Policies**
    - Dataset: OpenAI Gym (CartPole, LunarLander)
    - Method: PPO (Proximal Policy Optimization)
6. **Multimodal Foundation Model for Video Q&A**
    - Dataset: TVQA
    - Method: Vision Transformer + LLM fusion
7. **Instruction-Tuned LLMs for Task Planning**
    - Dataset: ALFRED task instructions
    - Method: Instruction fine-tuning of GPT/BERT with seq2seq decoding

---

**II. ML for Human-Machine Interaction / Action-Vision-Language**

8. **Vision-Language Navigation (VLN) in Indoor Environments**
    - Dataset: Room-to-Room (R2R)
    - Method: Seq2Seq + RL (Speaker-Follower model)
9. **Gesture-to-Command Mapping for Human-Robot Communication**
    - Dataset: Chalearn LAP Gesture Dataset
    - Method: CNN-LSTM sequence classification
10. **Emotion Recognition in Multimodal Conversations**

- Dataset: MELD
- Method: Multimodal fusion (Audio LSTM + Text BERT + Video CNN)

11. **Interactive QA with Video Context**

- Dataset: MovieQA, TVQA
- Method: VideoBERT + attention-based QA

12. **Speech-to-Action for Virtual Assistants**

- Dataset: LibriSpeech + synthetic action labels
- Method: Speech2Vec embeddings + Transformer decoder

13. **Cross-Modal Retrieval (Image ↔ Text)**

- Dataset: Flickr30K
- Method: Contrastive learning (CLIP-based encoders)

14. **Sign Language Recognition**

- Dataset: RWTH-PHOENIX-Weather 2014T
- Method: CNN + BiLSTM for sequence modeling

---

## III. ML for Robotics *

15. **Robot Grasp Detection using Deep RL**

- Dataset: Cornell Grasp Dataset, Dex-Net 2.0
- Method: Q-Learning with visual CNN input

16. **Sim-to-Real Transfer for Robotic Arm Manipulation**

- Dataset: MuJoCo simulations
- Method: Domain adaptation + RL fine-tuning

17. **Autonomous Drone Navigation in Cluttered Environments**

- Dataset: AirSim logs
- Method: PPO + CNN vision encoder

18. **Multi-Robot Coordination for Exploration**

- Dataset: Custom Gazebo simulations
- Method: Multi-Agent RL (MADDPG)

19. **Dexterous Manipulation with Imitation Learning**

- Dataset: D4RL Adroit Hand Dataset
- Method: Behavior Cloning + Diffusion Policy

20. **Robot Path Planning with Graph Neural Networks (GNNs)**

- Dataset: Grid-world simulations
- Method: GCN for path planning under uncertainty

21. **Human-Robot Collaboration for Assembly Tasks**

- Dataset: Assembly task demos (YCB dataset)
- Method: IL with human demonstration + RL refinement

---

## IV. ML for Agents (Gaming, Navigation, etc.)

22. **Deep RL Agent for Atari Games**

- Dataset: Atari Learning Environment (ALE)
- Method: DQN + Rainbow improvements

23. **Hierarchical RL for Minecraft (MineRL)**

- Dataset: MineRL ObtainDiamond
- Method: Hierarchical RL with options framework

24. **Language-Guided Game Agents**

- Dataset: BabyAI or TextWorld
- Method: LLM + RL hybrid for grounding instructions

25. **Multi-Agent Navigation in Crowd Environments**

- Dataset: Stanford Drone Dataset
- Method: Graph-based RL for collision avoidance

26. **StarCraft II Multi-Agent RL Challenge**

- Dataset: SMAC (StarCraft Multi-Agent Challenge)
- Method: QMIX + MADDPG for cooperation

27. **Soccer Simulation with Multi-Agent Reinforcement Learning**

- Dataset: RoboCup Simulation
- Method: Policy gradients for team coordination

28. **3D Maze Navigation with Vision-based RL**

- Dataset: DeepMind Lab
- Method: A3C with visual CNN

---

## V. ML for Quantitative Computing **

29. **Stock Price Forecasting with LSTM Networks**

- Dataset: Yahoo Finance / Quandl
- Method: LSTM for sequential time-series forecasting

30. **Volatility Prediction using GARCH + ML Hybrid**

- Dataset: NASDAQ/NYSE volatility indices
- Method: GARCH + Gradient Boosting

31. **Reinforcement Learning for Portfolio Optimization**

- Dataset: Bloomberg/WRDS stock datasets
- Method: Deep RL (DDPG, PPO)

32. **Crypto Price Forecasting with Transformers**

- Dataset: Binance/Kaggle crypto datasets
- Method: Time-series Transformers

33. **Algorithmic Trading with Deep Q-Learning**

- Dataset: Historical stock tick data
- Method: DQN with reward from profit/loss ratio

34. **Risk Prediction using Ensemble ML**

- Dataset: Lending Club Loan Dataset
- Method: XGBoost/LightGBM for credit risk

35. **High-Frequency Trading Order Book Prediction**

- Dataset: LOBSTER (Limit Order Book dataset)
- Method: CNN + LSTM hybrid

---

## VI. ML in Trustworthy Infrastructure or Safety Systems

36. **AI-based Intrusion Detection in Network Security**

- Dataset: CICIDS 2017
- Method: Autoencoders + Random Forest

37. **Anomaly Detection in Power Grid Systems**

- Dataset: IEEE Power Grid Dataset
- Method: GNN anomaly detection

38. **Autonomous Vehicle Safety with Adversarial Testing**

- Dataset: nuScenes, Waymo Open Dataset
- Method: Adversarial RL for robustness

39. **Bias Detection in Hiring Algorithms**

- Dataset: AI Fairness 360 (COMPAS, Adult dataset)
- Method: FairML, AIF360 with adversarial debiasing

40. **Privacy-Preserving ML with Federated Learning**

- Dataset: MNIST/CIFAR-10 (distributed partitions)
- Method: Federated Averaging (FedAvg) + secure aggregation

---

# 12 Selected ML Course Projects

## I. ML Foundation Models (RL / IL / LLM / VLM / AVM)

---

### Project 1: Fine-Tuning Large Language Models for Scientific Q&A

- **Objective:** Adapt pretrained LLMs (GPT/LLaMA) to answer biomedical/scientific questions with domain-specific accuracy.
- **Dataset:** PubMedQA, SQuAD biomedical subset.
- **Methodology:** LoRA fine-tuning + supervised learning on Q&A pairs, evaluation against baseline BERT QA models.
- **Expected Results:** Higher domain-specific accuracy and explainability vs. generic LLMs.
- **Evaluation Metrics:** Exact Match (EM), F1 score, BLEU.

---

### Project 2: Autonomous Video Modeling (AVM) for Human Action Recognition

- **Objective:** Develop a video transformer model to classify human actions from untrimmed video sequences.
- **Dataset:** Kinetics-700, UCF-101.
- **Methodology:** Video Transformer (TimeSformer/ViViT), pretrained on ImageNet frames, fine-tuned on action labels.
- **Expected Results:** Action recognition accuracy >85% on test set.
- **Evaluation Metrics:** Top-1 and Top-5 accuracy, confusion matrix.

---

## II. ML for Human-Machine Interaction (Action-Vision-Language)

---

### Project 3: Vision-Language Navigation (VLN) in Indoor Environments

- **Objective:** Train an RL-based agent to follow natural language instructions to navigate a simulated house.
- **Dataset:** Room-to-Room (R2R).
- **Methodology:** Speaker-Follower seq2seq model with attention; fine-tuned with RL reward for navigation success.
- **Expected Results:** Navigation success rate above 60% baseline.
- **Evaluation Metrics:** Navigation Error (meters), Success Rate, SPL (Success weighted by Path Length).

---

### Project 4: Emotion Recognition in Multimodal Conversations

- **Objective:** Build a multimodal classifier to detect emotions in dialogue (text, audio, video).
- **Dataset:** MELD (Multimodal EmotionLines).
- **Methodology:**
  - Text → BERT embeddings
  - Audio → LSTM on MFCC features
  - Video → CNN for facial features
  - Fusion → Attention-based multimodal fusion
- **Expected Results:** Emotion classification F1 > 70%.
- **Evaluation Metrics:** Accuracy, F1 score (macro-avg).

---

## III. ML for Robotics

---

### Project 5: Robot Grasp Detection using Deep RL

- **Objective:** Train a robotic arm to identify grasp points for objects using RL + vision.
- **Dataset:** Cornell Grasping Dataset, Dex-Net 2.0.
- **Methodology:** CNN for object detection + RL (Q-learning or PPO) for grasp execution policy.
- **Expected Results:** >80% successful grasps in simulation.
- **Evaluation Metrics:** Grasp success rate, mean grasp precision.

---

### Project 6: Autonomous Drone Navigation in Cluttered Environments

- **Objective:** Enable drones to autonomously navigate obstacle-rich spaces using visual RL.
- **Dataset:** AirSim drone simulation logs.
- **Methodology:** PPO with visual CNN encoder, domain randomization for robustness.
- **Expected Results:** Drone achieves >70% success in reaching targets in simulated maze.
- **Evaluation Metrics:** Success rate, average collision rate, path efficiency.

---

## IV. ML for Agents (Gaming / Navigation)

---

### Project 7: Deep RL Agent for Atari Games

- **Objective:** Train a DQN-based agent to play classic Atari games.
- **Dataset:** Atari Learning Environment (ALE).
- **Methodology:** Deep Q-Network with Rainbow improvements (dueling, prioritized replay).
- **Expected Results:** Human-level scores on Pong/Breakout.
- **Evaluation Metrics:** Average reward, score vs. human baseline.

---

### Project 8: Hierarchical RL for Minecraft (MineRL)

- **Objective:** Use HRL to solve long-horizon Minecraft tasks (e.g., ObtainDiamond).
- **Dataset:** MineRL dataset (expert trajectories).
- **Methodology:** Hierarchical RL with options framework (low-level mining/picking policies, high-level planning).
- **Expected Results:** Agent completing >30% of diamond tasks (above random baseline).
- **Evaluation Metrics:** Task completion rate, average reward.

---

## V. ML for Quantitative Computing

---

### Project 9: Stock Price Forecasting with LSTM Networks

- **Objective:** Predict next-day stock closing prices using sequential models.
- **Dataset:** Yahoo Finance daily stock data (AAPL, MSFT, TSLA).
- **Methodology:** LSTM with sliding window time-series input; comparison with ARIMA baseline.
- **Expected Results:** RMSE < 2% of stock price variance.
- **Evaluation Metrics:** RMSE, MAE, $R^2$.

---

### Project 10: Reinforcement Learning for Portfolio Optimization

- **Objective:** Build an RL agent to manage a portfolio for maximizing returns under risk constraints.
- **Dataset:** Bloomberg/WRDS historical stock returns.
- **Methodology:** Deep RL (DDPG or PPO) with Sharpe ratio as reward signal.
- **Expected Results:** Portfolio outperforming mean-variance optimization baseline.
- **Evaluation Metrics:** Sharpe ratio, annualized return, volatility.

---

## VI. ML in Trustworthy Infrastructure and Safety Systems

---

### Project 11: AI-based Intrusion Detection in Network Security

- **Objective:** Detect malicious network traffic in real-time with ML anomaly detection.
- **Dataset:** CICIDS 2017.
- **Methodology:** Autoencoder for anomaly detection + Random Forest for supervised classification.
- **Expected Results:** >95% detection of intrusions with low false positives.
- **Evaluation Metrics:** Accuracy, Precision-Recall, ROC-AUC.

---

### Project 12: Privacy-Preserving ML with Federated Learning

- **Objective:** Train models across distributed clients without centralizing sensitive data.
- **Dataset:** MNIST/CIFAR-10 partitioned across clients.
- **Methodology:** Federated Averaging (FedAvg) with secure aggregation; comparison vs. centralized baseline.
- **Expected Results:** Federated model achieves within 2% accuracy of centralized model.
- **Evaluation Metrics:** Test accuracy, communication cost, privacy leakage risk.

---